Algebra 2
Practice Test

Unit 2: Quadratic Functions

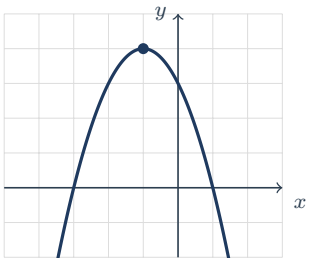
Instructions: Show all work. Give exact answers — simplest radical form, and $a + bi$ form for non-real numbers. No notes or calculator unless your teacher says so. **Total: 100 pts**

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Reading & Graphing Parabolas (20 pts)

1. (7 pts) The graph of a quadratic function f is shown. Read each feature from the graph.



Vertex: ☐

Axis of symmetry: ☐

Maximum or minimum? ☐

Its value: ☐

Zeros: ☐

Increasing on: ☐

Range: ☐

2. (3 pts) Which describes the graph of $y = -2(x - 3)^2 + 5$? **Letter:** ☐

(A) opens up; vertex (3, 5)

(B) opens down; vertex (−3, 5)

(C) opens down; vertex (3, 5)

(D) opens up; vertex (−3, 5)

3. (5 pts) For $f(x) = 2x^2 + 8x + 3$, use $x = -\dfrac{b}{2a}$ to find the axis of symmetry and the vertex.

Axis: ☐

Vertex: ☐

4. (5 pts) For $y = (x + 4)(x - 2)$, find the zeros, the axis of symmetry, and the vertex.

Zeros: ☐

Axis: ☐

Vertex: ☐

Part B — Solving by Factoring (20 pts)

Factor completely: take out the GCF first. To solve, get the equation equal to zero, factor, and set each factor equal to zero.

5. (8 pts) Factor completely.

   (a) $3x^2 - 12x - 36$

   (b) $x^2 - 81$

   (c) $x^2 - 10x + 25$

   (d) $3x^2 + 4x - 4$ (by grouping)

6. (4 pts) Solve $x^2 + 3x - 28 = 0$.

Solution:

7. (4 pts) Solve $2x^2 = 10x$.

Solution:

8. (4 pts) Solve $(x + 2)(x - 3) = 6$.

Solution:

Part C — Square Roots & Complex Numbers (20 pts)

Get the square alone before you take the root, and keep the $\pm$. Write non-real numbers in $a + bi$ form.

9. (3 pts) Solve $3x^2 + 5 = 65$.

Solution:

10. (4 pts) Solve $(x - 3)^2 = 12$.

Solution:

11. (4 pts) (a) Simplify $\sqrt{-72}$: ☐ (b) Solve $x^2 + 49 = 0$: ☐

12. (6 pts) Let $z_1 = 4 + 3i$ and $z_2 = 2 - 5i$. Write each answer in $a + bi$ form.

(a) $z_1 + z_2 =$ ☐ (b) $z_1 - z_2 =$ ☐

(c) $z_1 \cdot z_2$

$z_1 \cdot z_2 =$ ☐

13. (3 pts) $\sqrt{-4} \cdot \sqrt{-25} =$ ☐ (A) 10 (B) -10 (C) $10i$ (D) $-10i$

Part D — Completing the Square, Formula & Discriminant (20 pts)

14. (4 pts) Solve by completing the square: $x^2 + 8x - 5 = 0$

Solution: ☐

For 15 and 16, use the quadratic formula. Write the equation in standard form first.

15. (5 pts) Solve $2x^2 + 1 = 6x$.

Solution: ☐

16. (5 pts) Solve $x^2 + 6x + 13 = 0$.

Solution: ☐

17. (6 pts) For each equation, find the discriminant $b^2 - 4ac$. Then give the number and type of solutions (*two real*, *one repeated real*, or *two complex*) and the number of x-intercepts of its graph.

Equation	**Discriminant**	**Solutions**	**x-intercepts**
$x^2 - 6x + 9 = 0$	☐	☐	☐
$2x^2 + 3x + 4 = 0$	☐	☐	☐
$x^2 + 5x - 2 = 0$	☐	☐	☐

Part E — Systems & Modeling (20 pts)

Give each solution of a system as an ordered pair. Answer each model question with units.

18. (7 pts) Solve the system by substitution.

$$y = x^2 - 4x + 1 \qquad y = x - 3$$

Solutions: ____

19. (3 pts) How many points do the graphs of $y = x^2 + 1$ and $y = 2x$ have in common?

(A) none **Letter:** ____

(B) exactly one

(C) exactly two

(D) infinitely many

20. (10 pts) At the egg drop, an egg is launched straight up from the roof of the science building. Its height, in feet, t seconds after launch is

$$h(t) = -16t^2 + 48t + 64.$$

(a) (2 pts) How high is the roof? ____

(b) (4 pts) When does the egg reach its greatest height, and what is that height?

When: ____ Height: ____

(c) (4 pts) When does the egg hit the ground? Which solution do you reject, and why?

Lands at: ____

Rejected: ____